symGCAD / corpus benchmark

303 / 331 distinct original problems independently verified · 28 marked slow

All distinct problems are listed below, fastest verified solve first; unresolved problems follow. **Solve** and **check** are fresh-process wall seconds. **RAM** is maximum recorded resident MiB across solve and verification (†: verification RSS unavailable). **N** is the repetition count; three-run medians are preferred when available. **D**/**R** identifies debug/release builds. Cache tags: **C** = cold; **S** = shared checkpoints (may be warm); **?** = unrecorded. For incomplete rows, **TO**/**stop** is the longest retained observed attempt, not a solved time. Their RAM is the highest retained attempt RSS, which may come from an earlier run. The **baseline** columns are a separate, matched GCAD comparison: projection + lifting + merging only, with three fresh repetitions and identical orders. ± means two separate sign-branch solves, not the optimized full-split solve.

Matched ratio = symGCAD / GCAD: < 1 1–2 > 2 — = unavailable or censored; TO = native timeout. Ratios use two significant digits.

#	Original problem / representative full ID	IDs	Method · build	N	Solve s	Check s	RAM MiB	Cells	Matched Rust s (×R)	GCAD s
1	test_input_unmodified/solve/4	1	optimized 9025e71 R · ?	1	0.0412	0.0211	6.06	2	0.000076 **(×0.045)**	0.0017
2	finiteA_3_25_4_0__2_0_0_1_1_0_0_0_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.0413	0.0210	6.05	1	0.0011 **(×0.095)** ±	0.011
3	finiteA_3_21_4_0__2_0_1_0_1_0_0_0_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.0414	0.0216	6.07	5	0.00063 **(×0.053)** ±	0.012
4	test_ex44_B1_C2/solve/1	1	optimized 9025e71 R · ?	1	0.0427	0.0210	6.06	0	0.0077 **(×0.47)**	0.016
5	test_cylinder_3d/solve/1	1	optimized 9025e71 R · ?	1	0.0613	0.0211	6.02	1	0.00037 **(×0.099)**	0.0037
6	test_greedy_sotd_order/solve/2	1	optimized 9025e71 R · ?	1	0.0613	0.0210	6.04	7	0.00049 **(×0.11)**	0.0047
7	test_greedy_sotd_order/order/5	1	optimized 9025e71 R · ?	1	0.0613	0.0209	6.03	9	0.00067 **(×0.14)**	0.0049
8	test_empty_3d/solve/1	1	optimized 9025e71 R · ?	1	0.0613	0.0210	6.07	0	0.00013 **(×0.062)**	0.0020
9	test_no_real_roots/solve/1	1	optimized 9025e71 R · ?	1	0.0613	0.0210	4.46	1	0.00020 **(×0.16)**	0.0013
10	test_unbounded_neg/solve/1	1	optimized 9025e71 R · ?	1	0.0613	0.0209	4.46	1	0.000048 **(×0.036)**	0.0013
11	finiteE_1_1__3_0__3_0_0_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.0613	0.0209	6.06	1	0.00017 **(×0.043)** ±	0.0038
12	test_greedy_sotd_order/solve/1	1	optimized 9025e71 R · ?	1	0.0613	0.0210	6.02	9	0.00045 **(×0.095)**	0.0047
13	test_greedy_sotd_order/order/3	1	optimized 9025e71 R · ?	1	0.0613	0.0210	6.04	9	0.00046 **(×0.094)**	0.0049
14	test_x2_minus_y/solve/1	1	optimized 9025e71 R · ?	1	0.0613	0.0210	4.52	1	0.000065 **(×0.044)**	0.0015
15	test_quadrant/solve/1	1	optimized+H 9025e71 R · ?	1	0.0614	0.0210	4.57	1	0.000078 **(×0.051)**	0.0015
16	test_y2_minus_x/solve/1	1	optimized 9025e71 R · ?	1	0.0614	0.0210	4.54	1	0.00015 **(×0.088)**	0.0017
17	test_greedy_mods_order/order/1	1	optimized 9025e71 R · ?	1	0.0614	0.0211	6.03	6	0.00014 **(×0.045)**	0.0031
18	test_ex422_bubble/solve/1	1	optimized+H 9025e71 R · ?	1	0.0614	0.0211	6.04	2	0.00032 **(×0.092)**	0.0034
19	test_greedy_sotd_order/order/4	1	optimized 9025e71 R · ?	1	0.0614	0.0210	6.04	7	0.00046 **(×0.098)**	0.0047

#	Original problem / representative full ID	IDs	Method · build	N	Solve s	Check s	RAM MiB	Cells	Matched Rust s (×R)	GCAD s
20	finiteA_4_85_5_0__2_0_1_0_1_0_1_0_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.0614	0.0211	6.07	15	0.0026 **(×0.075)** ±	0.035
21	finiteA_4_154_5_0__0_2_0_1_1_0_0_1_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.0614	0.0211	6.05	8	0.0046 **(×0.095)** ±	0.048
22	test_x_y_z_minus_1/solve/1	1	optimized 9025e71 R · ?	1	0.0614	0.0211	6.05	2	0.00020 **(×0.039)**	0.0052
23	test_hypercube_2/solve/1	1	optimized 9025e71 R · ?	1	0.0614	0.0215	6.03	1	0.00016 **(×0.038)**	0.0043
24	finiteAx12_4_27_5_0__2_1_0_1_1_0_0_0_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.0615	0.0212	6.07	9	0.0026 **(×0.085)** ±	0.031
25	finiteAx12_4_154_5_0__0_2_0_1_1_0_0_1_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.0615	0.0211	6.07	8	0.0037 **(×0.11)** ±	0.034
26	finiteAx12_4_312_5_0__0_0_0_2_1_1_0_0_1/symbolic	1	monotone+H 9025e71 R · ?	1	0.0615	0.0211	6.06	9	0.0019 **(×0.075)** ±	0.026
27	finiteB_4_85_5_0__2_0_1_0_1_0_1_0_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.0615	0.0211	6.05	3	0.0015 **(×0.084)** ±	0.018
28	finiteAx12_3_26_4_0__0_2_0_1_1_0_0_0_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.0615	0.0211	6.07	8	0.00085 **(×0.063)** ±	0.013
29	test_input_unmodified/order/1	1	optimized 9025e71 R · ?	1	0.0615	0.0211	6.06	2	0.00013 **(×0.075)**	0.0017
30	test_hypercube_5/solve/1	1	optimized 9025e71 R · ?	1	0.0615	0.0211	6.04	2	0.00058 **(×0.098)**	0.0060
31	finiteA_4_293_5_0__2_0_1_0_0_1_0_0_1/symbolic	3	monotone+H 9025e71 R · ?	1	0.0615	0.0212	6.07	6	0.0014 **(×0.068)** ±	0.021
32	finiteA_4_89_5_0__2_0_0_1_1_0_1_0_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.0615	0.0211	6.05	8	0.0022 **(×0.069)** ±	0.032
33	finiteA_4_23_5_0__2_1_1_0_1_0_0_0_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.0615	0.0212	6.07	6	0.0010 **(×0.056)** ±	0.018
34	finiteE_2_10__3_0__0_2_0_1/symbolic	2	monotone+H 9025e71 R · ?	1	0.0615	0.0211	6.07	5	0.0012 **(×0.10)** ±	0.011
35	test_input_unmodified/order/2	1	optimized 9025e71 R · ?	1	0.0615	0.0211	6.04	2	0.000097 **(×0.047)**	0.0021
36	finiteA_4_297_5_0__2_0_0_1_0_1_0_0_1/symbolic	3	monotone+H 9025e71 R · ?	1	0.0615	0.0212	6.07	9	0.0039 **(×0.088)** ±	0.044
37	finiteAx12_4_297_5_0__2_0_0_1_0_1_0_0_1/symbolic	3	monotone+H 9025e71 R · ?	1	0.0615	0.0237	6.06	9	0.0033 **(×0.097)** ±	0.034
38	finiteA_4_278_5_0__0_2_1_0_1_0_0_0_1/symbolic	1	monotone+H 9025e71 R · ?	1	0.0615	0.0211	6.07	5	0.0010 **(×0.053)** ±	0.019
39	finiteB_4_197_5_0__2_0_1_0_0_0_1_1_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.0615	0.0211	6.07	13	0.0012 **(×0.062)** ±	0.019
40	finiteA_5_309_6_0__2_0_1_0_1_1_0_0_1/symbolic	2	monotone+H 9025e71 R · ?	1	0.0615	0.0212	6.06	42	0.40 **(×0.78)** ±	0.51
41	test_triangle2l_split/solve/3	1	optimized+H 9025e71 R · ?	1	0.0615	0.0211	6.05	6	0.0014 **(×0.086)**	0.016
42	finiteE_3_7__3_0__1_1_1_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.0615	0.0211	6.07	4	0.00066 **(×0.059)** ±	0.011
43	finiteA_4_27_5_0__2_1_0_1_1_0_0_0_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.0615	0.0211	6.06	9	0.0029 **(×0.082)** ±	0.036
44	test_triangle2l_split/order/1	2	optimized+H 9025e71 R · ?	1	0.0615	0.0211	6.05	4	0.0014 **(×0.081)**	0.017
45	test_ex411_box/solve/2	1	optimized+H 9025e71 R · ?	1	0.0615	0.0211	6.06	1	0.00018 **(×0.037)**	0.0048

#	Original problem / representative full ID	IDs	Method · build	N	Solve s	Check s	RAM MiB	Cells	Matched Rust s (×R)	GCAD s
46	finiteAx12_4_282_5_0__0_2_0_1_1_0_0_0_1/symbolic	3	monotone+H 9025e71 R · ?	1	0.0615	0.0211	6.06	36	0.0032 **(×0.094)** ±	0.034
47	test_ex44_B1_B4/solve/1	1	optimized 9025e71 R · ?	1	0.0615	0.0211	6.05	0	0.00091 **(×0.12)**	0.0074
48	test_input_unmodified/order/3	1	optimized 9025e71 R · ?	1	0.0615	0.0212	6.07	2	0.000099 **(×0.047)**	0.0021
49	finiteA_4_284_5_0__0_0_2_1_1_0_0_0_1/symbolic	3	monotone+H 9025e71 R · ?	1	0.0615	0.0212	6.05	5	0.0026 **(×0.080)** ±	0.032
50	finiteAx12_3_296_4_0__0_0_0_2_0_1_0_0_1/symbolic	2	monotone+H 9025e71 R · ?	1	0.0615	0.0211	6.07	6	0.00082 **(×0.061)** ±	0.014
51	finiteA_3_26_4_0__0_2_0_1_1_0_0_0_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.0615	0.0211	6.06	8	0.0018 **(×0.11)** ±	0.017
52	test_ex411_box/solve/1	1	optimized 9025e71 R · ?	1	0.0615	0.0211	6.09	1	0.00016 **(×0.034)**	0.0048
53	test_hypercube_3/solve/1	1	optimized 9025e71 R · ?	1	0.0615	0.0367	6.05	1	0.00019 **(×0.038)**	0.0050
54	finiteB_5_341_6_0__2_0_1_0_1_0_1_0_1/symbolic	3	monotone+H 9025e71 R · ?	1	0.0615	0.0211	6.05	11	0.0065 **(×0.085)** ±	0.077
55	finiteB_5_213_6_0__2_0_1_0_1_0_1_1_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.0615	0.0212	6.05	10	0.0028 **(×0.073)** ±	0.038
56	finiteAx12_3_296_4_0__0_0_0_2_0_1_0_0_1/fixed	2	monotone+H 9025e71 D · ?	1	0.0615	0.0211	12.1	1	0.00079 **(×0.096)** ±	0.0082
57	finiteA_5_87_6_0__2_1_1_0_1_0_1_0_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.0615	0.0212	6.06	28	0.0056 **(×0.088)** ±	0.064
58	finiteA_4_344_5_0__0_0_0_2_1_0_1_0_1/symbolic	3	monotone+H 9025e71 R · ?	1	0.0616	0.0212	6.05	10	0.0022 **(×0.071)** ±	0.030
59	finiteB_5_279_6_0__2_1_1_0_1_0_0_0_1/symbolic	2	monotone+H 9025e71 R · ?	1	0.0616	0.0213	6.43	18	0.0061 **(×0.11)** ±	0.054
60	finiteA_4_23_5_0__2_1_1_0_1_0_0_0_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0616	0.0211	12.1	6	0.00083 **(×0.059)** ±	0.014
61	finiteA_5_121_6_0__2_0_0_1_1_1_1_0_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.0616	0.0212	6.04	9	0.0038 **(×0.071)** ±	0.053
62	finiteA_2_17_3_0__2_0_0_0_1_0_0_0_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0616	0.0211	12.1	1	0.00017 **(×0.040)** ±	0.0043
63	finiteAx12_4_27_5_0__2_1_0_1_1_0_0_0_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0616	0.0211	12.1	1	0.0011 **(×0.063)** ±	0.018
64	finiteE_3_11__3_0__1_1_0_1/symbolic	2	monotone+H 9025e71 R · ?	1	0.0616	0.0211	6.07	8	0.0021 **(×0.093)** ±	0.023
65	finiteA_4_282_5_0__0_2_0_1_1_0_0_0_1/symbolic	3	monotone+H 9025e71 R · ?	1	0.0616	0.0211	6.04	36	0.0037 **(×0.093)** ±	0.040
66	test_melih_box/solve/3	1	optimized+H 9025e71 R · ?	1	0.0616	0.0212	6.43	3	0.0028 **(×0.081)**	0.034
67	finiteA_4_153_5_0__2_0_0_1_1_0_0_1_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0616	0.0212	12.1	1	0.0021 **(×0.088)** ±	0.024
68	finiteE_4_15__4_0__1_1_1_1/fixed	2	monotone+H 9025e71 D · ?	1	0.0616	0.0211	12.1	5	0.0021 **(×0.085)** ±	0.025
69	finiteA_5_310_6_0__0_2_1_0_1_1_0_0_1/symbolic	1	monotone+H 9025e71 R · ?	1	0.0616	0.0211	6.05	6	0.0024 **(×0.064)** ±	0.037
70	finiteA_3_296_4_0__0_0_0_2_0_1_0_0_1/fixed	2	monotone+H 9025e71 D · ?	1	0.0616	0.0212	12.1	1	0.00073 **(×0.077)** ±	0.0095
71	finiteB_3_21_4_0__2_0_1_0_1_0_0_0_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0616	0.0212	12.1	3	0.00030 **(×0.045)** ±	0.0067

#	Original problem / representative full ID	IDs	Method · build	N	Solve s	Check s	RAM MiB	Cells	Matched Rust s (×R)	GCAD s
72	finiteE_2_10__3_0__0_2_0_1/fixed	2	monotone+H 9025e71 D · ?	1	0.0616	0.0212	12.1	1	0.00070 **(×0.12)** ±	0.0059
73	finiteE_3_7__3_0__1_1_1_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0616	0.0212	12.1	4	0.00063 **(×0.070)** ±	0.0090
74	finiteA_3_276_4_0__0_0_2_0_1_0_0_0_1/fixed	2	monotone+H 9025e71 D · ?	1	0.0616	0.0212	12.1	4	0.00098 **(×0.12)** ±	0.0083
75	finiteA_4_312_5_0__0_0_0_2_1_1_0_0_1/symbolic	1	monotone+H 9025e71 R · ?	1	0.0616	0.0211	6.06	9	0.0022 **(×0.076)** ±	0.028
76	finiteA_3_21_4_0__2_0_1_0_1_0_0_0_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0616	0.0212	12.1	5	0.00095 **(×0.12)** ±	0.0080
77	finiteC_5_158_6_0__0_2_1_1_1_0_0_1_0/fixed	3	monotone+H 9025e71 D · ?	1	0.0616	0.0212	12.1	1	0.0047 **(×0.10)** ±	0.046
78	finiteC_5_214_10_0__0_6_1_0_1_0_1_1_0/fixed	5	monotone+H 9025e71 D · ?	1	0.0616	0.0212	12.1	1	0.019 **(×0.19)** ±	0.10
79	finiteA_3_25_4_0__2_0_0_1_1_0_0_0_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0616	0.0212	12.1	1	0.00091 **(×0.096)** ±	0.0095
80	finiteA_4_278_5_0__0_2_1_0_1_0_0_0_1/fixed	1	monotone+H 9025e71 D · ?	1	0.0616	0.0211	12.1	5	0.0010 **(×0.068)** ±	0.015
81	finiteF_4_15__4_0__1_1_1_1/fixed	1	monotone+H 9025e71 D · ?	1	0.0616	0.0212	12.1	1	0.0022 **(×0.085)** ±	0.026
82	test_ex412_box/solve/1	1	optimized 9025e71 R · ?	1	0.0616	0.0212	6.50	2	0.00022 **(×0.034)**	0.0065
83	finiteA_4_27_5_0__2_1_0_1_1_0_0_0_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0616	0.0212	12.1	1	0.0012 **(×0.065)** ±	0.019
84	finiteAx12_4_154_5_0__0_2_0_1_1_0_0_1_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0616	0.0212	12.1	1	0.0022 **(×0.097)** ±	0.023
85	finiteA_4_297_5_0__2_0_0_1_0_1_0_0_1/fixed	3	monotone+H 9025e71 D · ?	1	0.0616	0.0211	12.1	1	0.0017 **(×0.085)** ±	0.021
86	finiteE_2_9__3_0__2_0_0_1/fixed	1	monotone+H 9025e71 D · ?	1	0.0616	0.0213	12.1	1	0.00082 **(×0.14)** ±	0.0059
87	finiteA_5_171_6_0__2_1_0_1_0_1_0_1_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0616	0.0212	12.1	1	0.0044 **(×0.082)** ±	0.054
88	finiteA_3_276_4_0__0_0_2_0_1_0_0_0_1/symbolic	2	monotone+H 9025e71 R · ?	1	0.0616	0.0211	6.07	4	0.00075 **(×0.064)** ±	0.012
89	finiteC_5_227_6_0__2_1_0_0_0_1_1_1_0/fixed	3	monotone+H 9025e71 D · ?	1	0.0616	0.0213	12.1	1	0.0063 **(×0.11)** ±	0.056
90	finiteA_5_410_10_0__0_6_0_1_1_0_0_1_1/fixed	4	monotone+H 9025e71 D · ?	1	0.0616	0.0212	12.1	1	0.0085 **(×0.10)** ±	0.084
91	finiteAx12_4_297_5_0__2_0_0_1_0_1_0_0_1/fixed	3	monotone+H 9025e71 D · ?	1	0.0617	0.0212	12.1	1	0.0017 **(×0.081)** ±	0.020
92	finiteAx12_3_26_4_0__0_2_0_1_1_0_0_0_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0617	0.0211	12.1	1	0.00090 **(×0.11)** ±	0.0082
93	finiteAx12_4_170_5_0__0_2_0_1_0_1_0_1_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0617	0.0212	12.1	1	0.0020 **(×0.096)** ±	0.021
94	finiteA_4_282_5_0__0_2_0_1_1_0_0_0_1/fixed	3	monotone+H 9025e71 D · ?	1	0.0617	0.0211	12.1	1	0.0017 **(×0.082)** ±	0.021
95	finiteE_3_13__3_0__1_0_1_1/fixed	1	monotone+H 9025e71 D · ?	1	0.0617	0.0212	12.1	4	0.0013 **(×0.10)** ±	0.012
96	finiteA_5_286_6_0__0_2_1_1_1_0_0_0_1/fixed	3	monotone+H 9025e71 D · ?	1	0.0617	0.0212	27.5	6	0.0045 **(×0.091)** ±	0.049
97	finiteA_4_312_5_0__0_0_0_2_1_1_0_0_1/fixed	1	monotone+H 9025e71 D · ?	1	0.0617	0.0212	12.1	1	0.0018 **(×0.096)** ±	0.018

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98	finiteBx12_5_279_6_0__2_1_1_0_1_0_0_0_1/symbolic	2	monotone+H 9025e71 R · ?	1	0.0617	0.0213	6.47	18	0.0046 (×0.089) ±	0.052
99	finiteA_4_170_5_0__0_2_0_1_0_1_0_1_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0617	0.0212	12.1	1	0.0021 (×0.086) ±	0.024
100	finiteAx12_4_282_5_0__0_2_0_1_1_0_0_0_1/fixed	3	monotone+H 9025e71 D · ?	1	0.0617	0.0212	12.1	1	0.0017 (×0.083) ±	0.020
101	finiteAx12_4_312_5_0__0_0_0_2_1_1_0_0_1/fixed	1	monotone+H 9025e71 D · ?	1	0.0617	0.0211	12.1	1	0.0017 (×0.097) ±	0.018
102	finiteA_5_314_6_0__0_2_0_1_1_1_0_0_1/fixed	2	monotone+H 9025e71 D · ?	1	0.0617	0.0211	12.1	1	0.0054 (×0.098) ±	0.055
103	finiteA_4_29_5_0__2_0_1_1_1_0_0_0_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0617	0.0212	12.1	6	0.0017 (×0.091) ±	0.019
104	finiteC_5_31_6_0__2_1_1_1_1_0_0_0_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0617	0.0212	12.1	1	0.0041 (×0.098) ±	0.042
105	finiteA_4_29_5_0__2_0_1_1_1_0_0_0_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.0617	0.0212	6.05	6	0.0022 (×0.073) ±	0.030
106	finiteA_3_280_4_0__0_0_0_2_1_0_0_0_1/fixed	2	monotone+H 9025e71 D · ?	1	0.0617	0.0211	12.1	1	0.00079 (×0.082) ±	0.0096
107	finiteAx12_5_313_6_0__2_0_0_1_1_1_0_0_1/fixed	2	monotone+H 9025e71 D · ?	1	0.0617	0.0212	12.1	1	0.0064 (×0.11) ±	0.056
108	finiteA_5_310_6_0__0_2_1_0_1_1_0_0_1/fixed	1	monotone+H 9025e71 D · ?	1	0.0617	0.0213	27.6	6	0.0028 (×0.099) ±	0.029
109	test_ex422_bubble/solve/2	1	optimized+H 9025e71 R · ?	1	0.0617	0.0211	6.04	2	0.00031 (×0.079)	0.0039
110	finiteC_6_215_11_0__6_1_1_0_1_0_1_1_0/fixed	4	monotone+H 9025e71 D · ?	1	0.0617	0.0212	12.1	1	4.0 (×1.1) ±	3.5
111	finiteE_3_11__3_0__1_1_0_1/fixed	2	monotone+H 9025e71 D · ?	1	0.0617	0.0212	12.1	1	0.0011 (×0.096) ±	0.011
112	finiteA_3_26_4_0__0_2_0_1_1_0_0_0_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0618	0.0212	12.1	1	0.00084 (×0.090) ±	0.0094
113	finiteC_5_412_6_0__0_0_2_1_1_0_0_1_1/fixed	3	monotone+H 9025e71 D · ?	1	0.0618	0.0213	12.1	1	0.0030 (×0.085) ±	0.035
114	finiteAx12_5_314_6_0__0_2_0_1_1_1_0_0_1/fixed	2	monotone+H 9025e71 D · ?	1	0.0618	0.0213	12.1	1	0.0050 (×0.093) ±	0.054
115	finiteCx12_5_158_6_0__0_2_1_1_1_0_0_1_0/fixed	3	monotone+H 9025e71 D · ?	1	0.0618	0.0213	12.1	1	0.0047 (×0.100) ±	0.047
116	finiteA_5_31_6_0__2_1_1_1_1_0_0_0_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0618	0.0211	27.3	7	0.0038 (×0.084) ±	0.045
117	finiteA_5_313_6_0__2_0_0_1_1_1_0_0_1/fixed	2	monotone+H 9025e71 D · ?	1	0.0618	0.0213	12.1	1	0.0062 (×0.11) ±	0.055
118	finiteA_4_154_5_0__0_2_0_1_1_0_0_1_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0618	0.0215	12.1	1	0.0021 (×0.088) ±	0.024
119	test_ex44_T_B2/solve/1	1	optimized 9025e71 R · ?	1	0.0619	0.0211	6.06	31	0.0045 (×0.22)	0.021
120	finiteCx12_6_231_7_0__2_1_1_0_0_1_1_1_0/fixed	2	monotone+H 9025e71 D · ?	1	0.0619	0.0213	26.8	1	1.3 (×0.87) ±	1.5
121	finiteA_5_313_6_0__2_0_0_1_1_1_0_0_1/symbolic	2	monotone+H 9025e71 R · ?	1	0.0619	0.0215	11.4	10	0.034 (×0.25) ±	0.14
122	finiteA_5_155_6_0__2_1_0_1_1_0_0_1_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0619	0.0213	12.1	1	0.0046 (×0.088) ±	0.053
123	finiteAx12_5_171_6_0__2_1_0_1_0_1_0_1_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0619	0.0212	12.1	1	0.0042 (×0.083) ±	0.051

#	Original problem / representative full ID	IDs	Method · build	N	Solve s	Check s	RAM MiB	Cells	Matched Rust s (×R)	GCAD s
124	finiteA_4_284_5_0__0_0_2_1_1_0_0_0_1/fixed	3	monotone+H 9025e71 D · ?	1	0.0619	0.0212	12.1	5	0.0023 **(×0.10)** ±	0.022
125	finiteA_7_446_11_0__0_5_1_1_1_1_0_1_1/fixed	3	monotone+H 9025e71 D · ?	1	0.0620	0.0424	28.6	8	—	TO
126	finiteCx12_6_215_11_0__6_1_1_0_1_0_1_1_0/fixed	4	monotone+H 9025e71 D · ?	1	0.0620	0.0214	26.6	1	4.0 **(×1.1)** ±	3.6
127	finiteE_1_1__3_0__3_0_0_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0620	0.0211	16.4	1	0.00011 **(×0.042)** ±	0.0026
128	test_pentabox_exd/order/1	2	optimized+H 9025e71 R · ?	1	0.0621	0.0219	15.4	7	0.0077 **(×0.15)**	0.051
129	test_pentabox_exb/solve/3	1	optimized+H 9025e71 R · ?	1	0.0621	0.0216	15.4	3	0.080 **(×0.40)**	0.20
130	test_pentabox_exb/order/1	2	optimized+H 9025e71 R · ?	1	0.0622	0.0216	15.4	7	0.082 **(×0.35)**	0.23
131	test_gluza_ex33_trivial/solve/3	1	optimized+H 9025e71 R · ?	1	0.0622	0.0218	19.4	1	—	TO
132	test_gluza_ex33_trivial/order/1	2	optimized+H 9025e71 R · ?	1	0.0622	0.0218	19.4	0	—	TO
133	test_pentabox_exd/solve/3	1	optimized+H 9025e71 R · ?	1	0.0623	0.0218	15.4	4	0.0075 **(×0.16)**	0.048
134	finiteE_2_5__3_0__2_0_1_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0623	0.0212	12.1	3	0.00082 **(×0.14)** ±	0.0056
135	finiteA_5_157_6_0__2_0_1_1_1_0_0_1_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.0626	0.0212	6.05	6	0.0097 **(×0.11)** ±	0.090
136	finiteAx12_5_410_10_0__0_6_0_1_1_0_0_1_1/fixed	4	monotone+H 9025e71 D · ?	1	0.0627	0.0212	12.1	1	0.0084 **(×0.10)** ±	0.082
137	test_ex44_T_C1/solve/1	1	optimized 9025e71 R · ?	1	0.0629	0.0210	6.04	3	0.021 **(×0.51)**	0.041
138	finiteB_4_85_5_0__2_0_1_0_1_0_1_0_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0629	0.0211	12.1	3	0.0012 **(×0.086)** ±	0.014
139	test_ex44_B1_B2_B4/solve/1	1	optimized 9025e71 R · ?	1	0.0629	0.0211	6.06	0	0.0058 **(×0.29)**	0.020
140	finiteA_5_155_6_0__2_1_0_1_1_0_0_1_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.0631	0.0215	11.4	9	0.019 **(×0.16)** ±	0.12
141	finiteA_4_153_5_0__2_0_0_1_1_0_0_1_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.0632	0.0211	6.05	1	0.0023 **(×0.090)** ±	0.026
142	finiteA_5_316_10_0__0_0_6_1_1_1_0_0_1/fixed	5	monotone+H 9025e71 D · ?	1	0.0633	0.0213	27.7	6	0.018 **(×0.14)** ±	0.13
143	finiteAx12_5_155_6_0__2_1_0_1_1_0_0_1_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0635	0.0212	12.1	1	0.0047 **(×0.088)** ±	0.053
144	test_ex414_bnp6/solve/2	1	optimized 9025e71 R · ?	1	0.0635	0.0218	15.4	2	0.00057 **(×0.059)**	0.0096
145	finiteB_4_277_5_0__2_0_1_0_1_0_0_0_1/symbolic	3	monotone+H 9025e71 R · ?	1	0.0637	0.0211	6.05	3	0.0019 **(×0.064)** ±	0.029
146	finiteA_3_296_4_0__0_0_0_2_0_1_0_0_1/symbolic	2	monotone+H 9025e71 R · ?	1	0.0643	0.0244	6.05	6	0.0015 **(×0.092)** ±	0.016
147	test_cube/solve/1	1	optimized 9025e71 R · ?	1	0.0644	0.0210	6.04	1	0.00020 **(×0.055)**	0.0037
148	test_hypercube_4/solve/1	1	optimized 9025e71 R · ?	1	0.0651	0.0211	9.54	1	0.00071 **(×0.13)**	0.0056
149	test_ex415_bnp7/solve/2	1	optimized 9025e71 R · ?	1	0.0669	0.0219	19.4	10	0.0011 **(×0.052)**	0.020

#	Original problem / representative full ID	IDs	Method · build	N	Solve s	Check s	RAM MiB	Cells	Matched Rust s (×R)	GCAD s
150	test_boundary/solve/1	1	optimized 9025e71 R · ?	1	0.0814	0.0210	4.40	0	0.000051 **(×0.038)**	0.0013
151	test_massive_triangle/solve/2	1	optimized 9025e71 R · ?	1	0.0815	0.0211	6.09	4	0.00079 **(×0.11)**	0.0071
152	test_ex44_B1_B2_B3/solve/1	1	optimized 9025e71 R · ?	1	0.0817	0.0211	12.6	4	0.013 **(×0.67)**	0.019
153	finiteB_3_21_4_0__2_0_1_0_1_0_0_0_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.0817	0.0212	6.06	3	0.00037 **(×0.042)** ±	0.0088
154	finiteA_5_93_6_0__2_0_1_1_1_0_1_0_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.0817	0.0212	14.1	58	0.013 **(×0.11)** ±	0.11
155	finiteA_4_170_5_0__0_2_0_1_0_1_0_1_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.0818	0.0413	14.0	151	0.0052 **(×0.085)** ±	0.062
156	finiteC_6_95_7_0__2_1_1_1_1_0_1_0_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0818	0.0211	12.1	1	0.021 **(×0.12)** ±	0.17
157	finiteE_2_9__3_0__2_0_0_1/symbolic	1	monotone+H 9025e71 R · ?	1	0.0819	0.0211	6.07	1	0.00084 **(×0.12)** ±	0.0071
158	finiteB_4_277_5_0__2_0_1_0_1_0_0_0_1/fixed	3	monotone+H 9025e71 D · ?	1	0.0819	0.0212	26.6	3	0.0016 **(×0.075)** ±	0.022
159	finiteA_4_85_5_0__2_0_1_0_1_0_1_0_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0819	0.0212	26.8	15	0.0025 **(×0.12)** ±	0.022
160	finiteA_4_89_5_0__2_0_0_1_1_0_1_0_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0819	0.0212	26.8	8	0.0025 **(×0.11)** ±	0.024
161	finiteAx12_5_31_6_0__2_1_1_1_1_0_0_0_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0819	0.0211	27.3	7	0.0036 **(×0.080)** ±	0.045
162	finiteB_5_339_6_0__2_1_0_0_1_0_1_0_1/fixed	3	monotone+H 9025e71 D · ?	1	0.0819	0.0212	26.8	6	0.0034 **(×0.085)** ±	0.040
163	finiteCx12_6_438_7_0__0_2_1_0_1_1_0_1_1/fixed	1	monotone+H 9025e71 D · ?	1	0.0819	0.0213	12.1	1	0.041 **(×0.21)** ±	0.20
164	finiteC_6_470_10_0__0_5_1_0_1_0_1_1_1/fixed	6	monotone+H 9025e71 D · ?	1	0.0820	0.0213	27.3	7	0.14 **(×0.41)** ±	0.33
165	finiteB_5_213_6_0__2_0_1_0_1_0_1_1_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0820	0.0212	26.7	10	0.0025 **(×0.097)** ±	0.026
166	test_melih_box/order/1	2	optimized+H 9025e71 R · ?	1	0.0820	0.0213	6.43	1	0.0025 **(×0.073)**	0.035
167	finiteC_6_438_7_0__0_2_1_0_1_1_0_1_1/fixed	1	monotone+H 9025e71 D · ?	1	0.0820	0.0213	12.1	1	0.037 **(×0.20)** ±	0.19
168	finiteA_4_344_5_0__0_0_0_2_1_0_1_0_1/fixed	3	monotone+H 9025e71 D · ?	1	0.0820	0.0213	26.8	10	0.0024 **(×0.12)** ±	0.021
169	finiteB_6_343_7_0__2_1_1_0_1_0_1_0_1/fixed	2	monotone+H 9025e71 D · ?	1	0.0820	0.0228	27.0	15	0.0063 **(×0.092)** ±	0.069
170	finiteBx12_5_339_6_0__2_1_0_0_1_0_1_0_1/fixed	3	monotone+H 9025e71 D · ?	1	0.0820	0.0212	26.8	6	0.0034 **(×0.089)** ±	0.038
171	finiteCx12_7_247_10_0__4_1_1_0_1_1_1_1_0/fixed	3	monotone+H 9025e71 D · ?	1	0.0821	0.0212	27.1	1	—	TO
172	finiteAx12_5_301_6_0__2_0_1_1_0_1_0_0_1/fixed	3	monotone+H 9025e71 D · ?	1	0.0821	0.0213	26.9	8	0.0042 **(×0.090)** ±	0.047
173	finiteC_7_247_10_0__4_1_1_0_1_1_1_1_0/fixed	3	monotone+H 9025e71 D · ?	1	0.0821	0.0214	27.1	1	—	TO
174	finiteAx12_5_316_10_0__0_0_6_1_1_1_0_0_1/fixed	5	monotone+H 9025e71 D · ?	1	0.0821	0.0213	27.5	6	0.011 **(×0.15)** ±	0.073
175	finiteAx12_5_316_10_0__0_0_6_1_1_1_0_0_1/symbolic	5	monotone+H 9025e71 R · ?	1	0.0821	0.0424	14.6	249	—	TO

#	Original problem / representative full ID	IDs	Method · build	N	Solve s	Check s	RAM MiB	Cells	Matched Rust s (×R)	GCAD s
176	finiteA_5_412_6_0__0_0_2_1_1_0_0_1_1/fixed	3	monotone+H 9025e71 D · ?	1	0.0821	0.0212	27.5	6	0.014 **(×0.16)** ±	0.089
177	finiteA_6_190_7_0__0_2_1_1_1_1_0_1_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0822	0.0213	26.7	1	0.0095 **(×0.087)** ±	0.11
178	finiteB_5_341_6_0__2_0_1_0_1_0_1_0_1/fixed	3	monotone+H 9025e71 D · ?	1	0.0822	0.0212	26.8	11	0.0057 **(×0.11)** ±	0.051
179	finiteA_5_316_10_0__0_0_6_1_1_1_0_0_1/symbolic	5	monotone+H 9025e71 R · ?	1	0.0822	0.0416	14.7	151	—	TO
180	finiteA_5_314_6_0__0_2_0_1_1_1_0_0_1/symbolic	2	monotone+H 9025e71 R · ?	1	0.0822	0.0431	14.4	144	0.54 **(×0.68)** ±	0.80
181	finiteBx12_6_343_7_0__2_1_1_0_1_0_1_0_1/fixed	2	monotone+H 9025e71 D · ?	1	0.0823	0.0212	27.1	15	0.0065 **(×0.092)** ±	0.070
182	finiteA_6_414_10_0__0_5_1_1_1_0_0_1_1/fixed	4	monotone+H 9025e71 D · ?	1	0.0823	0.0213	28.4	7	3.3 **(×1.0)** ±	3.1
183	finiteA_5_376_6_0__0_0_0_2_1_1_1_0_1/fixed	3	monotone+H 9025e71 D · ?	1	0.0823	0.0212	26.9	12	0.0061 **(×0.13)** ±	0.048
184	test_ex412_box/solve/2	1	optimized 9025e71 R · ?	1	0.0824	0.0213	6.43	1	0.00023 **(×0.035)**	0.0065
185	finiteB_5_453_6_0__2_0_1_0_0_0_1_1_1/symbolic	1	monotone+H 9025e71 R · ?	1	0.0824	0.0615	15.0	524	0.0045 **(×0.094)** ±	0.048
186	finiteA_5_309_6_0__2_0_1_0_1_1_0_0_1/fixed	2	monotone+H 9025e71 D · ?	1	0.0824	0.0421	27.9	42	0.018 **(×0.30)** ±	0.058
187	test_ex413_pentagon/solve/2	1	optimized 9025e71 R · ?	1	0.0825	0.0219	19.0	3	0.00052 **(×0.037)**	0.014
188	finiteAx12_5_376_6_0__0_0_0_2_1_1_1_0_1/fixed	3	monotone+H 9025e71 D · ?	1	0.0825	0.0212	26.8	12	0.0058 **(×0.12)** ±	0.048
189	finiteA_5_301_6_0__2_0_1_1_0_1_0_0_1/fixed	3	monotone+H 9025e71 D · ?	1	0.0826	0.0211	26.9	8	0.0046 **(×0.095)** ±	0.048
190	finiteA_5_121_6_0__2_0_0_1_1_1_1_0_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0826	0.0211	27.1	9	0.0036 **(×0.086)** ±	0.042
191	test_pentabox_exb_symb/order/1	1	optimized 9025e71 R · ?	1	0.0830	0.0221	23.0	5	—	TO
192	finiteA_2_17_3_0__2_0_0_0_1_0_0_0_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.0831	0.0211	6.07	1	0.00019 **(×0.032)** ±	0.0058
193	finiteA_6_159_7_0__2_1_1_1_1_0_0_1_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0832	0.0212	27.5	7	0.029 **(×0.10)** ±	0.28
194	finiteA_5_87_6_0__2_1_1_0_1_0_1_0_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0832	0.0413	27.1	28	0.0045 **(×0.11)** ±	0.041
195	finiteE_2_5__3_0__2_0_1_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.0833	0.0212	6.07	3	0.00052 **(×0.073)** ±	0.0071
196	finiteC_6_476_10_0__0_0_5_1_1_0_1_1_1/fixed	4	monotone+H 9025e71 D · ?	1	0.0834	0.0211	27.5	7	0.038 **(×0.19)** ±	0.20
197	finiteA_5_157_6_0__2_0_1_1_1_0_0_1_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0836	0.0212	27.2	6	0.0049 **(×0.087)** ±	0.057
198	finiteB_4_197_5_0__2_0_1_0_0_0_1_1_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0836	0.0212	27.1	13	0.0012 **(×0.097)** ±	0.013
199	finiteA_6_318_7_0__0_2_1_1_1_1_0_0_1/fixed	2	monotone+H 9025e71 D · ?	1	0.0837	0.0213	28.5	7	0.72 **(×0.50)** ±	1.5
200	finiteAx12_4_170_5_0__0_2_0_1_0_1_0_1_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.0837	0.0413	11.0	151	0.0036 **(×0.094)** ±	0.038
201	finiteAx12_6_159_7_0__2_1_1_1_1_0_0_1_0/fixed	1	monotone+H 9025e71 D · ?	1	0.0837	0.0211	27.3	7	0.020 **(×0.13)** ±	0.16

#	Original problem / representative full ID	IDs	Method · build	N	Solve s	Check s	RAM MiB	Cells	Matched Rust s (×R)	GCAD s
202	finiteC_7_478_10_0__0_4_1_1_1_0_1_1_1/fixed	5	monotone+H 9025e71 D · ?	1	0.0838	0.0416	28.1	8	—	TO
203	finiteAx12_5_286_6_0__0_2_1_1_1_0_0_0_1/fixed	3	monotone+H 9025e71 D · ?	1	0.0839	0.0212	27.4	6	0.0043 **(×0.087)** ±	0.049
204	test_massive_triangle/solve/1	1	optimized 9025e71 R · ?	1	0.0846	0.0212	6.05	6	0.0010 **(×0.13)**	0.0076
205	finiteA_4_293_5_0__2_0_1_0_0_1_0_0_1/fixed	3	monotone+H 9025e71 D · ?	1	0.0846	0.0212	27.3	6	0.0016 **(×0.11)** ±	0.015
206	finiteAx12_6_318_7_0__0_2_1_1_1_1_0_0_1/fixed	2	monotone+H 9025e71 D · ?	1	0.0848	0.0213	27.8	7	0.67 **(×0.50)** ±	1.4
207	finiteAx12_5_313_6_0__2_0_0_1_1_1_0_0_1/symbolic	2	monotone+H 9025e71 R · ?	1	0.0895	0.0218	13.8	10	0.032 **(×0.28)** ±	0.11
208	test_hypercube_1/solve/1	1	optimized 9025e71 R · ?	1	0.102	0.0211	6.03	1	0.00013 **(×0.035)**	0.0037
209	finiteAx12_7_381_10_0__4_0_1_1_1_1_1_0_1/fixed	4	monotone+H 0a95bc2 R · C	3	0.102	0.0614	18.6	836	—	TO
210	finiteA_5_412_6_0__0_0_2_1_1_0_0_1_1/symbolic	3	monotone+H 9025e71 R · ?	1	0.102	0.0213	6.49	6	0.11 **(×0.22)** ±	0.49
211	finiteCx12_6_476_10_0__0_0_5_1_1_0_1_1_1/fixed	4	monotone+H 9025e71 D · ?	1	0.102	0.0212	27.3	7	0.040 **(×0.19)** ±	0.21
212	finiteAx12_6_414_10_0__0_5_1_1_1_0_0_1_1/fixed	4	monotone+H 9025e71 D · ?	1	0.102	0.0213	28.5	7	3.1 **(×1.0)** ±	3.0
213	finiteAx12_6_190_7_0__0_2_1_1_1_1_0_1_0/fixed	1	monotone+H 9025e71 D · ?	1	0.102	0.0212	26.7	1	0.0094 **(×0.085)** ±	0.11
214	finiteC_6_231_7_0__2_1_1_0_0_1_1_1_0/fixed	2	monotone+H 9025e71 D · ?	1	0.102	0.0212	26.7	1	1.2 **(×0.84)** ±	1.5
215	finiteA_6_125_7_0__2_0_1_1_1_1_1_0_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.102	0.0417	15.4	67	0.029 **(×0.13)** ±	0.22
216	finiteA_5_93_6_0__2_0_1_1_1_0_1_0_0/fixed	1	monotone+H 9025e71 D · ?	1	0.103	0.0624	27.7	58	0.0078 **(×0.14)** ±	0.058
217	finiteB_5_453_6_0__2_0_1_0_0_0_1_1_1/fixed	1	monotone+H 9025e71 D · ?	1	0.103	0.0419	27.6	77	0.0041 **(×0.12)** ±	0.035
218	finiteE_4_15__4_0__1_1_1_1/symbolic	2	monotone+H 9025e71 R · ?	1	0.103	0.0618	16.4	406	0.013 **(×0.13)** ±	0.10
219	test_ex44_B1_B2/solve/1	1	optimized 9025e71 R · ?	1	0.104	0.0211	6.04	1	0.0015 **(×0.18)**	0.0082
220	test_ex44_B1_C1/solve/1	1	optimized 9025e71 R · ?	1	0.107	0.0226	6.03	2	0.0051 **(×0.39)**	0.013
221	finiteAx12_5_155_6_0__2_1_0_1_1_0_0_1_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.107	0.0224	13.8	9	0.016 **(×0.18)** ±	0.089
222	finiteD_6_219_7_0__2_1_0_1_1_0_1_1_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.113	0.0417	16.3	262	4.5 **(×0.83)** ±	5.4
223	test_ex44_HT_HB2_HB3/solve/1	1	optimized 9025e71 R · ?	1	0.123	0.102	13.5	0	0.058 **(×0.48)**	0.12
224	finiteC_5_412_6_0__0_0_2_1_1_0_0_1_1/symbolic	3	monotone+H 9025e71 R · ?	1	0.123	0.0830	15.1	228	0.028 **(×0.28)** ±	0.10
225	test_ex415_bnp7/solve/1	1	optimized 9025e71 R · ?	1	0.123	0.0220	19.4	18	0.0010 **(×0.049)**	0.021
226	finiteA_6_317_7_0__2_0_1_1_1_1_0_0_1/fixed	2	monotone+H 9025e71 D · ?	1	0.123	0.0414	28.9	56	0.15 **(×0.43)** ±	0.36
227	finiteAx12_6_190_7_0__0_2_1_1_1_1_0_1_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.124	0.0422	19.4	130	0.055 **(×0.24)** ±	0.23

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228	finiteCx12_7_478_10_0__0_4_1_1_1_0_1_1_1/fixed	5	monotone+H 9025e71 D · ?	1	0.124	0.0414	28.6	8	—	TO
229	finiteAx12_5_410_10_0__0_6_0_1_1_0_0_1_1/symbolic	4	monotone+H 9025e71 R · ?	1	0.125	0.143	25.4	973	0.30 **(×0.35)** ±	0.85
230	failing_test_ex44_T_C1_B2/solve/1	1	optimized 9025e71 R · ?	1	0.125	0.0843	13.3	57	0.20 **(×1.3)**	0.16
231	finiteAx12_6_317_7_0__2_0_1_1_1_1_0_0_1/fixed	2	monotone+H 9025e71 D · ?	1	0.125	0.0416	28.9	56	0.14 **(×0.39)** ±	0.35
232	finiteAx12_5_314_6_0__0_2_0_1_1_1_0_0_1/symbolic	2	monotone+H 9025e71 R · ?	1	0.131	0.0888	15.5	144	0.49 **(×0.75)** ±	0.65
233	finiteAx12_7_446_11_0__0_5_1_1_1_1_0_1_1/fixed	3	monotone+H 9025e71 D · ?	1	0.142	0.0416	28.5	8	—	TO
234	test_ex414_bnp6/solve/1	1	optimized 9025e71 R · ?	1	0.143	0.0219	15.4	1	0.00039 **(×0.040)**	0.0099
235	finiteF_4_15__4_0__1_1_1_1/symbolic	1	monotone+H 9025e71 R · ?	1	0.143	0.144	19.4	624	0.071 **(×0.47)** ±	0.15
236	finiteC_5_227_6_0__2_1_0_0_0_1_1_1_0/symbolic	3	monotone+H 9025e71 R · ?	1	0.144	0.163	20.2	632	12 **(×0.92)** ±	13
237	failing_test_ex43/project/1	2	optimized 9025e71 R · ?	1	0.158	0.0211	12.6	10	0.079 **(×0.29)**	0.28
238	test_ex44_HB1_HB2_HB3/solve/1	1	optimized 9025e71 R · ?	1	0.163	0.147	14.1	0	0.033 **(×0.69)**	0.048
239	test_ex413_pentagon/solve/1	1	optimized 9025e71 R · ?	1	0.163	0.0219	19.0	5	0.00052 **(×0.037)**	0.014
240	finiteBx12_5_339_6_0__2_1_0_0_1_0_1_0_1/symbolic	3	monotone+H 9025e71 R · ?	1	0.164	0.123	25.0	954	0.027 **(×0.16)** ±	0.17
241	test_gluza_ex33/order/1	2	linear_negative 256527b R · C	3	0.164	0.0209	11.4	3	—	TO
242	finiteE_3_13__3_0__1_0_1_1/symbolic	1	monotone+H 9025e71 R · ?	1	0.166	0.0212	6.07	4	0.0020 **(×0.069)** ±	0.028
243	finiteD_6_219_7_0__2_1_0_1_1_0_1_1_0/fixed	1	monotone+H 9025e71 D · ?	1	0.166	0.123	29.7	262	0.061 **(×0.46)** ±	0.13
244	test_massless_planar_elliptic_symb/order/1	1	local+H e43a2c4 R · C	1	0.183	0.0211	6.00	12	—	TO
245	finiteBx12_6_343_7_0__2_1_1_0_1_0_1_0_1/symbolic	2	monotone+H 9025e71 R · ?	1	0.184	0.225	44.7	1963	0.075 **(×0.23)** ±	0.33
246	finiteA_6_125_7_0__2_0_1_1_1_1_1_0_0/fixed	1	monotone+H 9025e71 D · ?	1	0.184	0.103	29.1	67	0.014 **(×0.12)** ±	0.12
247	test_gluza_ex30/order/1	2	linear_negative 256527b R · C	3	0.185	0.0210	6.03	1	—	TO
248	finiteB_6_469_7_0__2_0_1_0_1_0_1_1_1/fixed	1	monotone+H 9025e71 D · ?	1	0.186	0.102	29.4	184	0.031 **(×0.22)** ±	0.14
249	finiteA_7_381_10_0__4_0_1_1_1_1_1_0_1/fixed	4	monotone+H 0a95bc2 R · C	3	0.186	0.0613	16.2	836	—	TO
250	test_pentabox_exd_symb/order/1	1	optimized 9025e71 R · ?	1	0.204	0.0425	23.0	152	—	TO
251	finiteA_5_410_10_0__0_6_0_1_1_0_0_1_1/symbolic	4	monotone+H 9025e71 R · ?	1	0.204	0.164	22.8	973	0.77 **(×0.35)** ±	2.2
252	finiteB_6_343_7_0__2_1_1_0_1_0_1_0_1/symbolic	2	monotone+H 9025e71 R · ?	1	0.205	0.285	46.5	2249	0.074 **(×0.24)** ±	0.30
253	finiteC_5_214_10_0__0_6_1_0_1_0_1_1_0/symbolic	5	monotone+H 9025e71 R · ?	1	0.206	0.164	20.2	632	—	TO

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254	finiteA_3_280_4_0__0_0_0_2_1_0_0_0_1/symbolic	2	monotone+H 9025e71 R · ?	1	0.223	0.0212	6.05	1	0.0010 **(×0.094)** ±	0.011
255	finiteBx12_5_279_6_0__2_1_1_0_1_0_0_0_1/fixed	2	monotone+H 9025e71 D · ?	1	0.223	0.0211	26.7	4	0.0024 **(×0.078)** ±	0.030
256	finiteCx12_5_158_6_0__0_2_1_1_1_0_0_1_0/symbolic	3	optimized d9488c7 R · ?	1	0.224	0.245	28.5	1273	0.74 **(×0.73)** ±	1.0
257	finiteA_6_190_7_0__0_2_1_1_1_1_0_1_0/symbolic	1	monotone+H 9025e71 R · ?	1	0.224	0.0648	19.4	130	0.098 **(×0.29)** ±	0.33
258	finiteD_7_375_13_0__7_1_1_0_1_1_1_0_1/fixed	4	monotone+H 9025e71 D · ?	1	0.224	0.124	31.1	296	3.3 **(×0.81)** ±	4.0
259	finiteD_6_373_7_0__2_0_1_0_1_1_1_0_1/fixed	2	monotone+H 9025e71 D · ?	1	0.225	0.123	30.3	214	0.83 **(×0.71)** ±	1.2
260	finiteBx12_7_471_13_0__7_1_1_0_1_0_1_1_1/fixed	4	monotone+H 9025e71 D · ?	1	0.226	0.164	31.0	245	0.067 **(×0.28)** ±	0.24
261	finiteA_5_345_6_0__2_0_0_1_1_0_1_0_1/fixed	4	monotone+H 9025e71 D · ?	1	0.226	0.185	29.7	309	0.033 **(×0.32)** ±	0.10
262	finiteB_7_471_13_0__7_1_1_0_1_0_1_1_1/fixed	4	monotone+H 9025e71 D · ?	1	0.227	0.164	31.0	245	0.057 **(×0.25)** ±	0.23
263	finiteD_7_251_12_0__6_1_0_1_1_1_1_1_0/fixed	6	monotone+H 9025e71 D · ?	1	0.227	0.246	31.5	364	—	partial TO
264	finiteB_6_469_7_0__2_0_1_0_1_0_1_1_1/symbolic	1	monotone+H 9025e71 R · ?	1	0.232	0.0417	14.8	184	0.60 **(×0.54)** ±	1.1
265	finiteB_5_279_6_0__2_1_1_0_1_0_0_0_1/fixed	2	monotone+H 9025e71 D · ?	1	0.244	0.0211	26.7	4	0.0023 **(×0.074)** ±	0.032
266	test_elliptic2l_physical/order/1	1	optimized+H 9025e71 R · ?	1	0.246	0.709	27.1	1196	1.8 **(×0.89)**	2.0
267	test_elliptic2l_physical/order/2	1	optimized+H 9025e71 R · ?	1	0.246	0.690	27.1	1196	1.2 **(×0.75)**	1.6
268	test_ex44_HT_C1_HB2/solve/1	1	optimized 9025e71 R · ?	1	0.247	0.366	19.7	0	0.16 **(×0.71)**	0.23
269	test_unbounded/solve/1	1	optimized+H 9025e71 R · ?	1	0.247	0.0210	8.81	1	0.000056 **(×0.053)**	0.0010
270	finiteAx12_5_376_6_0__0_0_0_2_1_1_1_0_1/symbolic	3	monotone+H 9025e71 R · ?	1	0.265	0.326	39.9	1744	21 **(×0.93)** ±	23
271	finiteD_6_373_7_0__2_0_1_0_1_1_1_0_1/symbolic	2	monotone+H 9025e71 R · ?	1	0.265	0.325	58.9	3096	—	TO
272	finiteA_5_286_6_0__0_2_1_1_1_0_0_0_1/symbolic	3	optimized d9488c7 R · ?	1	0.279	0.427	44.4	3709	1.6 **(×0.61)** ±	2.7
273	finiteC_5_158_6_0__0_2_1_1_1_0_0_1_0/symbolic	3	optimized d9488c7 R · ?	1	0.286	0.447	45.6	2197	1.2 **(×0.87)** ±	1.3
274	finiteB_5_339_6_0__2_1_0_0_1_0_1_0_1/symbolic	3	monotone+H 9025e71 R · ?	1	0.306	0.163	27.5	1079	0.11 **(×0.41)** ±	0.26
275	finiteA_6_377_10_0__5_0_0_1_1_1_1_0_1/fixed	4	monotone+H 9025e71 D · ?	1	0.307	0.225	31.5	413	1.1 **(×0.93)** ±	1.2
276	finiteAx12_6_377_10_0__5_0_0_1_1_1_1_0_1/fixed	4	monotone+H 9025e71 D · ?	1	0.329	0.225	31.5	413	0.92 **(×0.88)** ±	1.0
277	finiteA_5_345_6_0__2_0_0_1_1_0_1_0_1/symbolic	4	monotone+H 9025e71 R · ?	1	0.348	0.548	64.3	3706	0.38 **(×0.65)** ±	0.59
278	finiteAx12_5_286_6_0__0_2_1_1_1_0_0_0_1/symbolic	3	optimized d9488c7 R · ?	1	0.372	0.588	59.9	5489	3.1 **(×0.72)** ±	4.3
279	test_elliptic2l_physical/solve/3	1	optimized+H 9025e71 R · ?	1	0.468	0.770	39.5	1193	0.075 **(×0.49)**	0.15

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280	finiteAx12_5_301_6_0__2_0_1_1_0_1_0_0_1/symbolic	3	monotone+H 9025e71 R · ?	1	0.472	0.712	103	5462	3.1 (×0.76) ±	4.0
281	test_elliptic2l_physical/solve/4	1	optimized+H 9025e71 R · ?	1	0.530	0.811	56.4	3254	0.078 (×0.47)	0.16
282	finiteA_5_301_6_0__2_0_1_1_0_1_0_0_1/symbolic	3	monotone+H 9025e71 R · ?	1	0.592	1.02	145	7929	4.5 (×0.95) ±	4.8
283	finiteAx12_5_171_6_0__2_1_0_1_0_1_0_1_0/symbolic	1	monotone+H 9025e71 R · ?	1	1.18	2.14	252	15730	0.021 (×0.19) ±	0.11
284	test_elliptic2l_physical_symb/solve/2	1	optimized+H 9025e71 R · ?	1	1.36	2.55	196	5831	—	TO
285	finiteA_5_376_6_0__0_0_0_2_1_1_1_0_1/symbolic	3	monotone+H 9025e71 R · ?	1	1.68	2.38	225	12357	27 (×0.95) ±	28
286	finiteA_5_171_6_0__2_1_0_1_0_1_0_1_0/symbolic	1	monotone+H 9025e71 R · ?	1	1.81	2.63	252	15730	0.031 (×0.19) ±	0.16
287	finiteAx12_5_31_6_0__2_1_1_1_1_0_0_0_0/symbolic	1	monotone+H 9025e71 R · ?	1	2.32	4.15	590	35354	2.8 (×0.68) ±	4.2
288	finiteAx12_6_159_7_0__2_1_1_1_1_0_0_1_0/symbolic	1	monotone+H 9025e71 R · ?	1	2.85	5.26	635	35354	—	TO
289	test_ex_e5/order/1	2	optimized+H 9025e71 R · ?	1	4.95	11.2	170	0	9.3 (×2.7)	3.4
290	finiteA_5_31_6_0__2_1_1_1_1_0_0_0_0/symbolic	1	monotone+H 9025e71 R · ?	1	4.98	8.82	1235	74351	1.5 (×0.58) ±	2.6
291	test_ex_e5/solve/3	1	optimized+H 9025e71 R · ?	1	5.11	11.4	257	13266	9.4 (×2.1)	4.5
292	test_ex_e5/solve/4	1	optimized+H 9025e71 R · ?	1	5.34	14.0	373	16232	10 (×1.9)	5.2
293	test_ex_e5/solve/5	1	optimized+H 9025e71 R · ?	1	5.84	15.0	646	54026	10 (×1.2)	9.0
294	finiteA_6_159_7_0__2_1_1_1_1_0_0_1_0/symbolic	1	monotone+H 9025e71 R · ?	1	6.26	11.7	1329	74351	—	TO
295	test_massless_planar_elliptic/order/1	2	baseline 1d15d03 R · S	1	9.36	23.4	1630	131072	9.9 (×0.62)	16
296	test_massless_planar_elliptic/solve/3	1	baseline 1d15d03 R · S	1	9.37	23.6	1647	138678	9.8 (×0.62)	16
297	finiteC_6_438_7_0__0_2_1_0_1_1_0_1_1/symbolic	1	monotone+H 9025e71 R · ?	1	13.5	26.3	2357	123552	30 (×0.93) ±	32
298	finiteCx12_6_438_7_0__0_2_1_0_1_1_0_1_1/symbolic	1	monotone+H 9025e71 R · ?	1	21.4	32.7	2358	123552	44 (×0.93) ±	48
299	test_gluza_ex30/solve/3	1	bilinear_positive ff22294 R · C	3	22.6	16.9	744	80980	—	TO
300	test_elliptic2l_physical_symb/order/1	1	monotone−+H 1e29f04 R · C	1	33.9	73.1	6473	269061	—	TO
301	finiteC_5_31_6_0__2_1_1_1_1_0_0_0_0/symbolic	1	monotone+H 9025e71 R · ?	1	35.2	124	2286†	271030	1.1 (×0.83) ±	1.3
302	finiteC_6_95_7_0__2_1_1_1_1_0_1_0_0/symbolic	1	monotone+H 9025e71 R · ?	1	44.0	146	2527†	271030	—	TO
303	hanging_test_ex38/solve/1	1	optimized 9025e71 R · ?	1	56.8	80.8	805†	129064	55 (×3.1)	18
304	finiteA_6_317_7_0__2_0_1_1_1_1_0_0_1/symbolic	2	**slow / incomplete**	—	TO 3600	—	1364	—	—	TO
305	finiteA_6_318_7_0__0_2_1_1_1_1_0_0_1/symbolic	2	**slow / incomplete**	—	TO 3600	—	1413	—	—	TO
306	finiteA_6_377_10_0__5_0_0_1_1_1_1_0_1/symbolic	4	**slow / running**	—	TO 60.0	—	208	—	—	TO
307	finiteA_6_414_10_0__0_5_1_1_1_0_0_1_1/symbolic	4	**slow / running**	—	TO 60.0	—	970	—	—	TO

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308	finiteA_7_381_10_0__4_0_1_1_1_1_1_0_1/symbolic	4	slow / running	—	TO 60.0	—	467	—	—	TO
309	finiteA_7_446_11_0__0_5_1_1_1_1_0_1_1/symbolic	3	slow / incomplete	—	TO 3600	—	1944	—	—	TO
310	finiteAx12_6_317_7_0__2_0_1_1_1_1_0_0_1/symbolic	2	slow / running	—	TO 60.0	—	106	—	—	TO
311	finiteAx12_6_318_7_0__0_2_1_1_1_1_0_0_1/symbolic	2	slow / pending	—	TO 60.0	—	135	—	—	TO
312	finiteAx12_6_377_10_0__5_0_0_1_1_1_1_0_1/symbolic	4	slow / pending	—	TO 60.0	—	164	—	—	TO
313	finiteAx12_6_414_10_0__0_5_1_1_1_0_0_1_1/symbolic	4	slow / pending	—	TO 60.0	—	142	—	—	TO
314	finiteAx12_7_381_10_0__4_0_1_1_1_1_1_0_1/symbolic	4	slow / pending	—	TO 60.0	—	485	—	—	TO
315	finiteAx12_7_446_11_0__0_5_1_1_1_1_0_1_1/symbolic	3	slow / incomplete	—	TO 3600	—	1022	—	—	TO
316	finiteB_7_471_13_0__7_1_1_0_1_0_1_1_1/symbolic	4	slow / pending	—	TO 60.0	—	3738	—	—	TO
317	finiteBx12_7_471_13_0__7_1_1_0_1_0_1_1_1/symbolic	4	slow / pending	—	TO 60.0	—	2782	—	—	TO
318	finiteC_6_215_11_0__6_1_1_0_1_0_1_1_0/symbolic	4	slow / pending	—	TO 60.0	—	186	—	—	TO
319	finiteC_6_231_7_0__2_1_1_0_0_1_1_1_0/symbolic	2	slow / pending	—	TO 60.0	—	1877	—	—	TO
320	finiteC_6_470_10_0__0_5_1_0_1_0_1_1_1/symbolic	6	slow / pending	—	TO 60.0	—	242	—	—	TO
321	finiteC_6_476_10_0__0_0_5_1_1_0_1_1_1/symbolic	4	slow / pending	—	TO 60.0	—	2025	—	—	TO
322	finiteC_7_247_10_0__4_1_1_0_1_1_1_1_0/symbolic	3	slow / pending	—	TO 60.0	—	1031	—	—	TO
323	finiteC_7_478_10_0__0_4_1_1_1_0_1_1_1/symbolic	5	slow / pending	—	TO 60.0	—	322	—	—	TO
324	finiteCx12_6_215_11_0__6_1_1_0_1_0_1_1_0/symbolic	4	slow / pending	—	TO 60.0	—	92	—	—	TO
325	finiteCx12_6_231_7_0__2_1_1_0_0_1_1_1_0/symbolic	2	slow / pending	—	TO 60.0	—	2347	—	—	TO
326	finiteCx12_6_476_10_0__0_0_5_1_1_0_1_1_1/symbolic	4	slow / pending	—	TO 60.0	—	2416	—	—	TO
327	finiteCx12_7_247_10_0__4_1_1_0_1_1_1_1_0/symbolic	3	slow / pending	—	TO 60.0	—	776	—	—	TO
328	finiteCx12_7_478_10_0__0_4_1_1_1_0_1_1_1/symbolic	5	slow / pending	—	TO 60.0	—	308	—	—	TO
329	finiteD_7_251_12_0__6_1_0_1_1_1_1_1_0/symbolic	6	slow / pending	—	TO 60.0	—	163	—	—	TO
330	finiteD_7_375_13_0__7_1_1_0_1_1_1_0_1/symbolic	4	slow / pending	—	TO 60.0	—	101	—	—	TO
331	test_gluza_ex33/solve/3	1	slow / pending	—	TO 60.0	—	179	—	—	TO

Measurement notes and implementation

Scope and comparability

The table contains all 331 exact polynomial/domain/order equivalence groups, including fixed and symbolic ZZ examples, core examples, and disabled upstream fixtures. “IDs” counts the retained original aliases; the accompanying `data.json` records every alias, input identity, source report and executable hash.

A row is complete only when the full original problem has an independently verified result. Completing its positive and negative branches separately does not automatically certify the full split problem. The current audit has 303 verified groups and 28 unresolved groups. The unresolved set is marked slow and assigned full one-hour campaigns; this snapshot must be regenerated as those runs finish. Missing times and cells are not zeros, and timeouts do not produce finite speedup ratios.

The first one-hour wave (10 October 2026) ended with four timeouts: symbolic A317, A318, A446 and Ax12-A446 each ran 3600.024 seconds without exporting a result. The audited queue then had four further runs active and twenty pending. These attempts add no completion credit; the frozen queue snapshot and ledger hashes are in the JSON.

Measurements prefer verified three-repeat controls, otherwise the fastest accepted original-problem run. Builds are labelled. Cache tags mean cold (C), shared and possibly warm (S), or unrecorded (?). Solve times include in-run search, preprocessing and export. Earlier order discovery is separate; Gluza30 positive used a disclosed 300.026-second search. Verification has its own column.

Baseline ratios use the frozen `9025e71` release and original native GCAD, with identical inputs, orders, merge settings, CPU assignment and paired limits. For two sign branches, displayed times sum their separate three-repeat median phases; ratios compare those matched sums. They do not compare the optimized solve column with an unrelated native time. All 483 native-completing branches matched raw cell counts and independently verified; 96 native timeouts remain explicit. Geometric mean ratios are 0.15 for matched phases and 0.062 for whole processes.

New reference comparisons use at most max(300 s, 3 × completed symGCAD solve-wall median) per native repetition, subject to shorter explicit and campaign limits. Qualification runs use a 300-second ceiling before Rust timings are known. Native timeouts have no exact ratio; the historical baseline above retains its recorded limits.

Machine and resource policy

AMD EPYC 9754 128-Core Processor, Linux 6.18.45, x86-64. The task budget is 16 CPU workers and 64 GiB aggregate memory; each measured solver process uses one CPU. Historical per-worker memory ceilings range from 4 to 32 GiB and are retained in the source ledgers. RAM is measured resident memory, not the configured ceiling. Three historical supplemental verification runs retain elapsed time but no RSS; their RAM entries carry † and report the available solve RSS only. Concurrent campaigns may share the host, so single-run differences are not presented as precise speedups. The ongoing slow campaign uses bounded process supervision, durable attempt records and independent proof verification.

Supplemental sanity check. Fixed A446 passed two fresh rational points in each of its eight exported cells (seed 20261010): 16 exact original-F evaluations and 112 strict-constraint checks, with zero failures or unchecked cells. Boundaries were recomputed at each fresh prefix. This separate check is outside benchmark timings and supplies no coverage-proof credit; full points and counterexamples from a deliberately wrong sign label are retained in `reference/SUPPLEMENTAL_SANITY.md`.

What changed

Exact algebra and reuse. Rust uses Symbolica for polynomial arithmetic, factorization, resultants and rational root work. Canonical factors, deterministic algorithms and bounded caches avoid repeated work. Certified sum-variable preprocessing and primitive parameter-linear factors address measured algebra bottlenecks.

Smaller valid decompositions. Positive-domain certificates, homogeneous charts, conditional quadratic monotone reduction and strict linear/bilinear identities can avoid irrelevant projection factors or restore whole fibers from a smaller verified base. Fixed A381/Ax381 and the negative Gluza cases now have independently verified compact decompositions. Every specialized rule carries exact replayable evidence; ordinary projection remains a fallback.

Bounded local and ordering work. Local projection supports positive-ray certificates, finite witness stocks, strict Bézout pair separation and explicit resource stops. MCTS preserves variable groups and fixed-order obligations, uses bounded trial tiers and resumable journals, and admits winners only after fresh completion. Optional local progress scoring changes promotion priority, not mathematical proof acceptance.

Reproduce and inspect

The Typst source and frozen `data.json` render without private literature or original proof files. Regeneration from retained evidence uses the Rust example `benchmark_report` its source records the selection rules and rejects missing identities or changed audited ledgers. Run `nix develop -c typst compile benchmark/benchmark.typ benchmark/benchmark.pdf`. Full provenance remains in the JSON snapshot and the repository’s benchmark/reference notes.

Variable ordering / bounded MCTS

Search mechanics · 1 of 2

The objective is the fastest completed ordering found within a budget. symGCAD implements Monte Carlo tree search (MCTS) over variable-elimination prefixes. Exact solves measure candidates; heuristic scores guide where to spend the next trial. This search supplies no certificate that its chosen ordering is globally optimal.

01 · Seed
Given order + inexpensive polynomial heuristics

02 · Explore
UCT selects a prefix; a rollout fills the remaining order

03 · Measure
Bounded exact solves; promising timeouts get longer trials

04 · Select
Fastest completed candidate; otherwise a restart suggestion

Why order matters

Each projection eliminates a variable and forms discriminants, leading coefficients and pairwise resultants. An unfortunate early elimination can increase degrees, term counts and coefficient sizes dramatically, making every later step expensive. Lifting cost also depends on the resulting root boundaries and number of cells. Consequently, the final choice uses completed solve time, including lifting.

The tree and its finite depth

A node represents an elimination prefix; an edge chooses the next legal variable. With *n* variables, a complete path has depth *n*. Variable groups constrain the legal branches. An explicit fixed order bypasses MCTS, and a homogeneous scale anchor must retain its required position.

Public orders are written in **lifting order**: an order `[x, y, z]` means eliminate z, then y, then x. The tree uses this elimination direction internally. For example, `[z, y]` fixes the first two eliminations, while a rollout fills in the remaining variables.

The implementation has **no separate depth-k lookahead setting**. It does not stop after k eliminations, choose one variable and restart the search at the next level. Its limits are elapsed time and scheduler iterations; each evaluated candidate is a complete ordering, although its algebra worker can time out partway through.

Inexpensive starting orders

The initial portfolio contains the given order and deduplicated MODS-, T1-, Brown- and sparsity-based rankings. These seeds use the original input features, avoiding a full projection for each seed. Expansion and rollout priorities likewise use a fixed input ranking; they are not recomputed from every projected polynomial set.

Selection, rollout and feedback

Unexpanded legal choices are tried in heuristic priority order. Once all children exist, UCT combines mean reward with an exploration bonus and a small, decaying preference for the heuristic ranking. The default score for a visited child is

$$Q_i + \sqrt{2}\sqrt{\frac{\ln N}{n_i}} + \frac{0.1}{(r_i+1)(n_i+1)}$$

Here Q is mean reward, N the parent visit count, n the child visit count and r its zero-based heuristic rank. Unvisited children take priority. A rollout follows the cheapest-ranked legal choice 80% of the time; otherwise it samples a legal choice uniformly. Seed 0 is the default.

Completed solves earn the largest rewards, with faster solves scoring higher. A timed-out global-projection trial can earn a smaller reward from observed factor, term, degree and coefficient-size costs. These observations predict difficulty; they do not estimate how much of the solution set has been proved.

Successive time budgets

Fresh orders start with a one-second worker limit. Timed-out candidates can be promoted to ten seconds, then sixty seconds. Promotion normally follows eight fresh trials, or happens sooner when all legal orders have been seen. Every fourth promotion favors a deeper tier so longer trials are not starved.

Killable worker processes enforce deadlines even inside an expensive resultant or factorization. Each trial is clipped to the remaining search budget. The optimized engine can also abandon prefixes with extreme observed term growth and avoid repeating them during that search campaign.

What wins

Only a completed exact solve can become `best_order`. The comparison uses measured worker wall time, with fewer cells breaking an exact timing tie. With no completion, `suggested_order` is a heuristic restart candidate. The public solve command then runs the selected order with the remaining campaign time and exports that solve's evidence.

Variable ordering / use and evidence

Budgets, reproducibility and interpretation · 2 of 2

Enable the search

Add these settings to a problem TOML, keeping its variables, polynomial expressions and domain constraints. Leave the explicit `order` field unset.

```toml
[solver]
engine = "optimized"
ordering = "mcts"
seed = 0
max_rollouts = 256
search_time_secs = 300
checkpoint_dir = "artifacts/checkpoints"

[limits]
wall_time_secs = 3600
memory_mib = 4096
workers = 1
```

`symgcad order problem.toml` runs the bounded search and reports its history. `symgcad solve problem.toml` includes the search, then solves the selected ordering. `symgcad verify result.json` checks the exported decomposition separately. MCTS is opt-in; the default ordering is `given`.

How the limits interact

Control	Effective behavior
Search wall time	At most min(300 s, 20% of campaign); an explicit search limit can lower it.
`max_rollouts`	256 scheduler iterations by default; promotions and revisits also consume iterations.
Candidate tiers	1 s ⟶ 10 s ⟶ 60 s, clipped to remaining search time.
Final solve	Receives the remaining campaign wall budget after search.

A one-hour campaign therefore allows up to five minutes for search; a 300-second campaign allows up to sixty seconds. The iteration ceiling does not promise 256 distinct exact solves. Process launch and scheduling can add small timing overheads.

Reuse and restart

Compatible trials share completed exact algebra and projection-prefix caches. An atomic journal preserves trial history, random-generator state through replay, tree visits and promotion state. Context checks bind the problem, domains, groups, solver settings and executable identity. Restored completion claims must pass a fresh bounded solve before they can win; the journal never substitutes for proof.

Soundness and the meaning of “best”

Choosing or abandoning an ordering changes search effort only. It cannot delete a solution region, omit an ordinary $F = 0$ sheet or justify a projection reduction. The selected final solve does not use the search’s heuristic prefix-abandonment rule. Its complete status still requires exhaustive algebraic construction, with an independently checked proof required for benchmark completion credit.

“Best” means the fastest completed candidate observed under the recorded limits, engine and cache conditions. It is not an exhaustive comparison of all permutations. A favorable partial score, a timeout or an old journal entry cannot establish a solved result. If the final solve also fails to finish, the problem remains incomplete.

What the measurements establish

One controlled experiment tested the optional `search_local_progress = true` policy on a five-dimensional positive Gluza30 base using the local engine. The option uses polynomial inventory and refinement growth to rank timed-out candidates for promotion. It is disabled by default and changes neither UCT rewards nor the completed-winner rule.

Promotion policy	Verified	Median solve
Existing scoring	0 / 3	313.62 s stop
Local inventory scoring	3 / 3	65.10 s

These paired fresh-process runs used the same release executable, seed 0, cold checkpoints, a sixty-second search budget, one CPU and 4 GiB per worker. Enabled runs produced 15,389 cells; median separate verification took 2.50 seconds. The reported solve time includes search. The unfinished controls do not yield an exact speed ratio, and this experiment compares two MCTS promotion policies, not MCTS against a heuristic-only solver.

This is evidence of benefit on one reduced problem. The earlier multi-problem ordering study did not establish an overall MCTS advantage; the new result does not establish a corpus-wide improvement or explain the matched-baseline speedup. Search overhead and shared-cache effects must remain visible in comparisons.

Implementation and audit trail

`src/search.rs`: tree policy, rewards, tier scheduling and prefix abandonment.
`src/ordering.rs`: inexpensive seed portfolio and group conventions.
`src/main.rs`: process trials, budgets and final solve.
`src/search/checkpoint.rs`: bounded journal and restart validation.
`docs/USAGE.md`: configuration and command usage.
`reference/LOCAL_SEARCH_PROMOTION_BENCHMARK.md`: paired experiment, limits, source identities and retained evidence.